CHAPTER 3

Function limits

S21, S23, S24, S26

Before you start · Function limits

Without a calculator, in six minutes. Answer briefly. Mobilize the sign, factorization and the then choose the targeted pick-up indicated under the questions.

1. Factoring $x^2 - 4$.
2. Give the sign of $\frac{x-1}{x+2}$ on $]-2;1[$.
3. Simplify $\sqrt{x^2}$ for $x < 0$.
4. What is the limit of the sequence $\frac{1}{n+1}$?
5. Determine the dominant term of 3 $x^3 - x$.
6. To give a prohibited value for $\frac{1}{x-2}$.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Develop (x-2)(x+2) to check. **2.** Make two lines of signs and then quotient them. **3.** Test x=-3 and then generalize; R28. **4.** Resuming the unit limits of sequences and band around 0. **5.** Compare the degrees before any simplification. **6.** Cancel denominator before calculation. Questions R serve as support with the teacher, even if their collective ritual is planned later.

Objectives

Calculating limits at one point and infinitely; lifting an indetermination; interpreting a Asymptote.

Course notes

Sense of a limit

$f(x) \to \ell$ where $x \to +\infty$ means that $f(x)$ can be made as close as desired to ℓ for all x large enough. a, x of the estate, close to a and distinct from a. The limits to the left and right are marked $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. When the domain has points on both sides of aa bilateral limit exists if and only if Two lateral limits exist and are equal. At one end of the domain, the limit of the only accessible side.

Graphical interpretation

If $f(x) \to \ell \in \mathbb{R}$ where $x \to +\infty$ or $-\infty$, right $y = \ell$ is a horizontal asymptote in that direction. $f(x) \to \pm\infty$ where $x \to a$ one side, the right $x = a$ is an asymptote vertical.

A curve can cut a horizontal asymptote. The limit does not necessarily give the position of the curve relative to the asymptote: study the sign of $f(x) - \ell$.

References and operations

In $+\infty$, $x^m \to +\infty$ for $m \geq 1$, $\frac{1}{x^m} \to 0$, $\sqrt{x} \to +\infty$. In $-\infty$, the sign x^m depends on the parity of m. The rules of sum, product, quotient and comparison are those of the sequences, with the same forms Indeterminate.

Example. For $f(x) = \frac{2x+1}{x-3}$, write $f(x) = 2 + \frac{7}{x-3}$. Limitations in $\pm\infty$ 2 in 3 $^-$, $f \to -\infty$; in 3^+, $f \to +\infty$. Asymptotes are $x = 3$ and $y = 2$.

Maritalization. For $x > 0$, $\sqrt{x^2+4x} - x = \frac{4}{\sqrt{1+\frac{4}{x}}+1} \to 2$. For $x < 0$, attention: $\sqrt{x^2} = |x| = -x$.

Exponential and Comparative Growth

$$\lim_{x\to+\infty} e^x = +\infty, \quad \lim_{x\to-\infty} e^x = 0, \quad \lim_{x\to+\infty} \frac{e^x}{x^m} = +\infty \quad (m \in \mathbb{N}).$$

Deduced from it $x^m e^{-x} \to 0$ in $+\infty$.

Demonstration of the limits of the exponential. The study of $e^x - x - 1$ given $e^x \geq 1 + x$. By comparison, $e^x \to +\infty$ in $+\infty$. By posing $t = -x \to +\infty$, $e^x = \frac{1}{e^t} \to 0$.

Demonstration of comparative growth. For the integer $k \geq 0$ and $x \geq 0$, $e^x \geq \frac{x^k}{k!}$. For $k = 0$, $e^x \geq 1$. If inequality is acquired by order k, the derivative of $e^x - \frac{x^{k+1}}{(k+1)!}$ is $e^x - \frac{x^k}{k!} \geq 0$; its value in 0 is positive. Heredity follows. By choosing $k = m + 1$, $\frac{e^x}{x^m} \geq \frac{x}{(m+1)!} \to +\infty$.

Choose a transformation

Polynomial quotient: factor the greatest powers. Root difference: quantity
quotient near a zero: factorize and study the signs.
power: compared growth, after variable change if necessary.

Comparison and composition: conditions of employment

If $g(x) \leq f(x) \leq h(x)$ near the point or infinity studied, and if g, h have the same real limit ℓ, then $f(x) \to \ell$ (Theorem of the gendarmes) $g \leq f$ and $g \to +\infty$, then $f \to +\infty$. For a limit $-\infty$, on uses an increase by a function tending to $-\infty$. If $u(x) \to a$ and v is continuous in a, then $v(u(x)) \to v(a)$, when the compound is defined. For example $\sqrt{1+\frac{1}{x}} \to 1$ in $+\infty$. For a limit infinite of the external function, check which side $u(x)$ Approach the bound: $\frac{1}{u(x)}$ Not the same behaviour according to the sign of $u(x)$.

Read a limit, choose a transformation

3.1 The approach side is part of the question

First use: S23.

Example Solved · Vertical Asymptote

For $f(x) = 1 + \frac{1}{x-2}$, the domain is $\mathbb{R} \setminus \{2\}$. When $x \to 2^-$ the denominator is negative and tend towards 0, so $f(x) \to -\infty$. When $x \to 2^+$, $f(x) \to +\infty$. The right $x = 2$ is asymptote vertical; no value $f(2)$ exists. Infinity, $f(x) \to 1$.

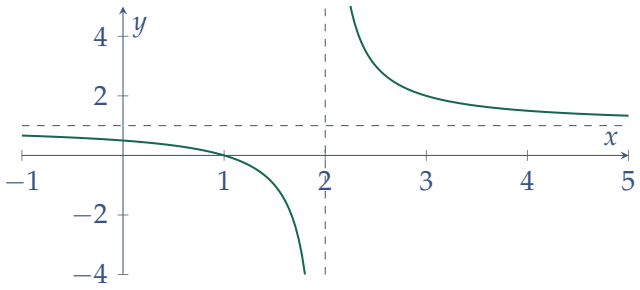


Figure 3.1. Find the two lateral boundaries on the drawing: the branches are not connected.

A1 · Immediate application Determining the limits of $g(x) = 3 - \frac{2}{x+1}$ in -1^-, -1^+, $-\infty$ and $+\infty$. Indicate asymptotes.

3.2 Undetermined form requires calculation

First use: S24.

Example Solved · The sign hidden in a root

For $x < 0$, $\sqrt{x^2} = |x| = -x$. Thus, in $-\infty$,

$$\frac{\sqrt{x^2+1}}{x} = \frac{|x|\sqrt{1+1/x^2}}{x} = -\sqrt{1+1/x^2} \longrightarrow -1.$$

In $+\infty$, the same expression tends to 1. Omitting the absolute value would produce a limit positive for a negative expression.

A2 · Guidance which disappears Calculate in $+\infty$ the limits of $\frac{2x^2+1}{x^2-x}$, $\sqrt{x^2+4x} - x$ and $x\left(\sqrt{1+1/x} - 1\right)$. For the first, it is suggested to divide by x^2 ; choose alone for others.

A3 · Build Examples Build three limit quotients of form $0/0$ in 0, of which the limits are 0, 1 and $+\infty$. Specify the approach side if necessary.

3.3 Comparative growth and variable change

First use: S26; logarithm after S49.

Ownership · exponential and power · course results

For the integer $m \geq 1$, $e^x / x^m \to +\infty$ and $x^m e^{-x} \to 0$ in $+\infty$. To transport these results, identify the new variable and its limit.

Example Solved · Carrying a Result

When $x \to -\infty$, pose $t = -2x$ gives $t \to +\infty$ and $x^2 e^{2x} = \frac{1}{4} t^2 e^{-t} \to 0$. After S49, substitution $t = 1/x$ in $x \ln x$ for $x \to 0^+$ gives $-(\ln t)/t \to 0$. Logarithm n
Not before his study.

A4 · Deferred application After S26, calculate $\lim_{x \to +\infty} (x^2 + 3x) e^{-x}$ and $\lim_{x \to -\infty} x e^x$. To say why formally replace x Infinity does not constitute justification.

3.4 An asymptote may be crossed

First use: S24.

Definition · Horizontal Asymptote

If $f(x) - \ell \to 0$ to one of the infinite, $y = \ell$ is an asymptote to this infinity.
a vertical distance that tends to 0; it does not prohibit any intersection at a finite distance.

Example Solved · Two Behaviours

On $\mathbb{R}$, $p(x) = 1 + 1/(1 + x^2)$ remains strictly above $y = 1$. The function $q(x) = 1 + x/(1 + x^2)$ cross that same right in $x = 0$, while tending towards 1 to the two infinites.
For q, the sign $q(x) - 1$ is the x.

A5 · Refute and Explain A student says that the curve of q Can't cut his asymptote.
the coordinates of the point of intersection and the signs on both sides. Is it compatible with the Two limits?

A6 · Choice of strategy Determine a, b so that $h(x) = \frac{ax+b}{x-1}$ have for horizontal asymptote $y = 2$ and verifies $h(0) = 3$. Then study the lateral limits in 1 and the position of the curve in relation to $y = 2$.

Wait a minute at the courthouse. For each of the expressions of the "Methods" application, announce first
This prevents us from concluding with the boundary operations and then explaining the role of the transformation.

Review postponed · S49 and then S81 Resume variable change with $\ln x / x$; reclassify limits by mixing logarithms, exponentials and roots.

Research activity

A curve close to a right · 30 min.

The function is considered f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. We note $\mathcal{C}_f$ its curve in a coordinate system of the plan.

1. Show that, for all $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.
2. Determine the limits of f in $-\infty$, 1^-, 1^+ and in $+\infty$.
3. Study the sign of $f(x) - 3$, then specify the position of $\mathcal{C}_f$ in relation to the right of equation $y = 3$.
4. Determining a Real $A > 1$ such as, for all $x > A$, $|f(x) - 3| < 0{,}01$.
5. The Curve $\mathcal{C}_f$ Does it cut its horizontal asymptote? Justify and explain why a window too wide graphic can give a misleading impression.

Progressive exercises

E1 · Direct application. Determine $\lim_{x \to +\infty} \dfrac{5x^3 - x}{2x^3 + 7}$ justifying.

E2 · Application and justification. Let f the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits in 2^- in 2^+, in $-\infty$ and in $+\infty$.

E3 · Method to be organised. Determine $\lim\limits_{x \to 1} \dfrac{x^2 - 1}{x - 1}$ specifying why the simplification used is valid in the vicinity of 1.

E4 · Choice of strategy. Determine $\lim_{x \to +\infty} x^3 e^{-2x}$. Cite the comparative growth result used.

E5 · Synthesis or demonstration. Determine $\lim\limits_{x \to -\infty} \left(\sqrt{x^2 + 1} + x\right)$. Justify in particular the treatment of $\sqrt{x^2}$.

Work in the manual or annal

Barbazo, p. 96, Exercises 26 and 29. In S23: Exercise 29, to connect boundaries and asymptotes. At home After S24: Exercise 26, to construct three counter-examples to an abusive calculation rule on limits.

Duty at home

DM3 · A family of curves · 45 min.

For every real a, the function is considered f_a defined by $f_a(x) = \frac{ax^2+1}{x^2+1}$. We note $\mathcal{C}_a$ its curve representative.

1. Determine the definition set of f_a, then its limits in $-\infty$ and in $+\infty$.
2. Show that $f_a(x) - a = \frac{1-a}{x^2+1}$. Depending on the value of a, specify the $\mathcal{C}_a$ in relation to the right of equation $y = a$.
3. Studying variations in f_a distinguishing cases $a < 1$, $a = 1$ and $a > 1$. Clarification of possible extremema and where they are affected.
4. Take $a = 3$. Resolve, for $x \geq 0$, the inequitable $|f_3(x) - 3| \leq 0{,}02$.

QCM A · lateral boundaries and transformations

After S24; QCM B after S26.

For each question, only one proposal is accurate.
consult the correction.

1. $\lim_{x\to 2^-} \frac{1}{x-2}$ is valid for:

A. $+\infty$ B. $-\infty$

C. 0 D. 2

2. For $x < 0$, $\sqrt{x^2}$ is valid for:

A. x B. x^2

C. $-x$ D. $\pm x$

3. $\lim_{x\to+\infty} \frac{3x^2-x}{x^2+4}$ is valid for:

A. 0 B. $+\infty$

C. -1 D. 3

4. $\lim_{x\to 1} \frac{x^2-1}{x-1}$ is valid for:

A. 2 B. 0

C. $+\infty$ D. Does not exist

5. For $f(x) = 1 + \frac{\sin x}{x}$, $x > 0$, the right $y = 1$:

A. is not asymptote B. is asymptote without intersection

C. is asymptote crossed an infinity of times D. is tangent at all points

QCM B · Comparative growth and tool choice

For each question, only one proposal is accurate.
consult the correction.

6. $\lim_{x\to+\infty} x^2e^{-x}$ is valid for:

A. $+\infty$ B. 0

C. 1 D. $-\infty$

7. $\lim_{x\to-\infty} xe^x$ is valid for:

A. 0 B. $+\infty$

C. $-\infty$ D. 1

8. The appropriate tool for $\sqrt{x^2+2x} - x$ in $+\infty$ is:

A. the binomial B. VIT

C. a diversion D. Conjugation

9. If $f(x) \to 0$ and $g(x) \to +\infty$the product:

A. Always tends to 0 B. Always tend towards $+\infty$

C. requires an additional study D. Is impossible

10. If $f(x) \geq x^2$ for $x \geq 1$, then:

A. $f(x)/x^2 \to 1$ B. $f(x) \to +\infty$

C. f is increasing D. $f(x) - x^2 \to 0$

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. A function never cuts its horizontal asymptote.

2. A finite limit in a requires that $f(a)$ be defined.

3. The symbol ∞/∞ gives a limit equal to 1.